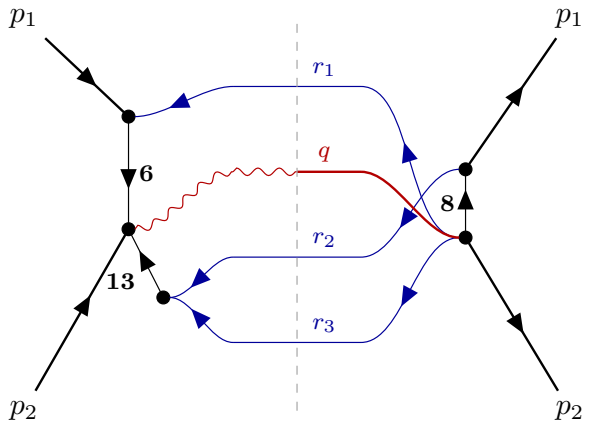
# Top sectors as cut diagrams

Amplitude (left) × conjugate (right) for $p_1 p_2 \to V(q)\, r_1 r_2 r_3$, glued along the cut (dashed): $r_1, r_2, r_3$ (blue), $q$ (red). The rapidity condition $-q\cdot p_1 + u\, q\cdot p_2 = 0$ of every family is not a line: the $q$ line is drawn wavy up to the cut ($q^2 = m^2$) and as a double line after it (the rapidity condition). Internal lines carry their DD number. Arrows: momentum flow with all legs incoming ($p_1 + p_2 + r_1 + r_2 + r_3 + q = 0$): in the amplitude every line carries the sum of the incoming momenta on its far side, e.g. line 6 carries $p_1 + r_1$; the conjugate is the mirror image, so $p_1, p_2$ leave on the right and $r_i, q$ flow through the cut from right to left (the physical final-state momenta $-r_i$ from left to right). **Regular propagators** (DD numbers, all legs incoming): $6 = (p_1 + r_1)^2$; $7 = (p_2 + r_1)^2$; $8 = (p_1 + r_2)^2$; $9 = (p_2 + r_2)^2$; $10 = (p_1 + r_3)^2$; $11 = (p_2 + r_3)^2$; $12 = (r_1 + r_2)^2$; $13 = (r_2 + r_3)^2$; $14 = (r_1 + r_3)^2$; $15 = (p_1 + r_1 + r_2)^2$; $16 = (p_2 + r_1 + r_2)^2$; $17 = (p_1 + r_1 + r_3)^2$; $18 = (p_2 + r_1 + r_3)^2$; $19 = (p_1 + r_2 + r_3)^2$; $20 = (p_2 + r_2 + r_3)^2$; $21 = (r_1 + r_2 + r_3)^2$; $22 = (p_1 + r_1 + r_2 + r_3)^2$; $23 = (p_2 + r_1 + r_2 + r_3)^2$; $24 = (p_1 + p_2 + r_1)^2$; $25 = (p_1 + p_2 + r_2)^2$; $26 = (p_1 + p_2 + r_3)^2$; $27 = (p_1 + p_2 + r_1 + r_2)^2$; $28 = (p_1 + p_2 + r_1 + r_3)^2$; $29 = (p_1 + p_2 + r_2 + r_3)^2$.




**FAM[55]** (topo55), sector with 3 of 7 lines

| | | |
|---|---|---|
| **6** | $(p_1 + r_1)^2$ | amplitude |
| **13** | $(r_2 + r_3)^2$ | amplitude |
| **8** | $(p_1 + r_2)^2$ | conjugate |

pinched: $7 = (p_2 + r_1)^2$, $14 = (r_1 + r_3)^2$, $18 = (p_2 + r_1 + r_3)^2$, $22 = (p_1 + r_1 + r_2 + r_3)^2$